

Multi-Platform E2E Testing & Bug Matrix

QA MASTER PLAN

Plexuspharmaco Europe Quality Assurance & Verification Suite

TARGET APPLICATION	TARGET PLATFORMS	TEST ENGINE	BUG CHECKLIST ITEMS
plexuspharmaco-eu.vercel.app	iOS, Android, Windows, macOS	Playwright E2E & WebKit/Chromium	105 Verified Test Cases

20

APPLE IOS & SAFARI

20

ANDROID CHROME/OS

20

WINDOWS & DESKTOP

20

LAYOUT & UX

25

SECURITY & SPEED

Executive Scope: This document outlines the comprehensive cross-platform End-to-End (E2E) testing framework and automated verification checklist for **Plexuspharmaco Europe**. It targets full UI/UX integrity, functional reliability, regulatory compliance presentation, and technical stability across Apple WebKit, Google Chromium, and Windows environments.

1. Cross-Platform Playwright Configuration Matrix

Deploy the following configuration in `playwright.config.ts` to execute tests in parallel across mobile and desktop environments:

```
import { defineConfig, devices } from '@playwright/test';

export default defineConfig({
  testDir: './e2e',
  use: {
    baseURL: 'https://plexuspharmaco-eu.vercel.app',
    trace: 'on-first-retry',
    screenshot: 'only-on-failure',
  },
  projects: [
    { name: 'Windows Chrome', use: { ...devices['Desktop Chrome'] } },
    { name: 'macOS Safari', use: { ...devices['Desktop Safari'] } },
    { name: 'iOS Safari (iPhone 14)', use: { ...devices['iPhone 14 Pro'] } },
    { name: 'Android Chrome (Pixel 7)', use: { ...devices['Pixel 7'] } },
  ],
});
```

2. Complete 105-Point Bug Inspection Matrix

CATEGORY 1: APPLE IOS & WEBKIT SAFARI BUGS (ITEMS 1 – 20)

#	SEVERITY	BUG / INSPECTION PARAMETER	TECHNICAL ROOT CAUSE & VERIFICATION POINT
01	HIGH	iOS 100vh Viewport Cutoff	Safari bottom navigation bars conceal lower CTAs. Requires 100dvh instead of static 100vh.
02	MEDIUM	Input Auto-Zoom on iOS	Form inputs with font sizes under 16px trigger automatic screen magnification on iPhone focus.
03	MEDIUM	Sticky Header Jitter	Rapid momentum scrolling causes sticky/fixed headers to flicker on WebKit engines.

CATEGORY 1: APPLE IOS & WEBKIT SAFARI BUGS (ITEMS 1 – 20)

#	SEVERITY	BUG / INSPECTION PARAMETER	TECHNICAL ROOT CAUSE & VERIFICATION POINT
04	HIGH	Notch & Dynamic Island Cutoff	Missing viewport-fit=cover and env(safe-area-inset-top) causing text clipping at the top.
05	LOW	Rubber-Band Flash	Over-scroll bounce reveals a jarring white background mismatch behind dark theme wrappers.
06	LOW	Native Date Picker Distortion	iOS wheel picker default styles misaligning within custom styled form elements.
07	MEDIUM	Backdrop Blur Invalidation	Glassmorphism navbars failing to render blur without -webkit-backdrop-filter prefix.
08	MEDIUM	300ms Click Latency	Missing touch-action: manipulation introducing tap delay on interactive buttons.
09	LOW	Sticky Hover State on Tap	CSS :hover styles persisting after touch release on mobile Safari touchscreens.
10	LOW	Font Smoothing Discrepancy	Retina screens rendering thin typography poorly without -webkit-font-smoothing: antialiased.
11	MEDIUM	Nested Momentum Scroll Lock	Missing -webkit-overflow-scrolling: touch causing stiff scrolling in modal dialogs.
12	MEDIUM	SVG Icon Scaling Collapse	Vector logos collapsing or distorting when explicit viewBox or height attributes are omitted.
13	LOW	Auto Number Telephone Linking	Safari auto-styling numbers/codes as blue phone links. Requires telephone=no meta tag.
14	HIGH	Background Video Autoplay Failure	Videos failing to auto-start due to missing playsinline, muted, or low-power mode blocks.
15	HIGH	Virtual Keyboard Modal Displacement	Virtual keyboard triggering pushes fixed-position modal overlays out of the visible screen frame.
16	LOW	Unintended Card Text Highlighting	User swipe interactions accidentally triggering blue text selection over clickable cards.
17	LOW	Default Inner Input Shadow	iOS applying default gray inner shadow on text fields unless -webkit-appearance: none is set.
18	MEDIUM	3D Transform Z-Index Glitch	CSS transform: translate3d breaking standard stacking context on WebKit renders.
19	LOW	Native Media Control Badges	Default QuickTime play overlay masking custom video preview placeholders.
20	MEDIUM	Smart Invert Contrast Inversion	iOS accessibility dark inversion making transparent corporate brand logos invisible.

CATEGORY 2: ANDROID & MOBILE CHROMIUM BUGS (ITEMS 21 – 40)

#	SEVERITY	BUG / INSPECTION PARAMETER	TECHNICAL ROOT CAUSE & VERIFICATION POINT
21	MEDIUM	Dynamic Address Bar Reflow	Address bar collapsing/expanding triggers visual layout shifts and re-renders in flex layouts.
22	LOW	Tap Highlight Overlay Box	Grey/blue rectangular flash on touch tap; fix with <code>-webkit-tap-highlight-color: transparent</code> .
23	HIGH	Keyboard Push Sticky Footer	Android software keyboard pushing fixed bottom floating contact buttons into the middle screen.
24	MEDIUM	Hardware Acceleration Jitter	Complex CSS box-shadows causing noticeable frame drops on mid-range Android devices.
25	MEDIUM	Hardware Back Button Closes Page	Android back key navigates away rather than intercepting and closing open inquiry modals.
26	LOW	Roboto Baseline Alignment	System font fallback causing slight vertical misalignment in rounded status pills and badges.
27	LOW	AMOLED Wide-Gamut Oversaturation	Unprofiled CSS colors appearing overly saturated on Samsung OLED displays without sRGB color space.
28	LOW	Autofill Yellow Background	Chrome auto-fill credentials overriding custom input theme with default yellow/blue fills.
29	MEDIUM	Pull-to-Refresh Gesture Conflict	Horizontal swiping across product carousels triggering browser pull-to-refresh reload.
30	HIGH	Touch Target Size Under 48px	Close navigation icons and links failing minimum accessibility touch target size guidelines.
31	MEDIUM	Lazy-Loaded Layout Thrashing	Images loading asynchronously without aspect-ratio tags, shifting text down after loading.
32	MEDIUM	Modern WebP/AVIF Fallback Loss	Legacy embedded Android WebViews showing blank images when next-gen formats lack fallbacks.
33	LOW	Native Select Dropdown Truncation	Android select dialogs truncating long European pharmaceutical classification names.
34	LOW	Missing Web App Manifest	Add-to-Homescreen failing to display branded icon due to incomplete <code>manifest.webmanifest</code> .
35	LOW	Multi-Touch Pinch Disruption	Interactive maps or interactive diagrams breaking when zoomed with two fingers simultaneously.
36	HIGH	OS Font Scale 200% Breakage	Large system accessibility text sizes overflowing fixed-height card containers.
37	MEDIUM	Landscape Orientation Re-render	Screen rotation failing to trigger responsive grid column adjustments on Android tablets.
38	MEDIUM	Touch Drag Gesture Sticking	Slider cards getting stuck midway between slides due to unhandled <code>touchcancel</code> events.
39	LOW	Status Bar Theme Color Missing	Missing <code><meta name="theme-color"></code> leaving the browser navigation chrome mismatched.
40		Cookie Banner Mobile Blocking	

CATEGORY 2: ANDROID & MOBILE CHROMIUM BUGS (ITEMS 21 – 40)

#	SEVERITY	BUG / INSPECTION PARAMETER	TECHNICAL ROOT CAUSE & VERIFICATION POINT
	HIGH		GDPR cookie modal covering the entire mobile screen with unreachable accept buttons.

CATEGORY 3: WINDOWS & CROSS-BROWSER DESKTOP BUGS (ITEMS 41 – 60)

#	SEVERITY	BUG / INSPECTION PARAMETER	TECHNICAL ROOT CAUSE & VERIFICATION POINT
41	MEDIUM	Scrollbar Layout Shift (15px)	Windows Chrome/Edge non-overlay scrollbar insertion causes modal opening layout shudder.
42	LOW	ClearType Font Pixelation	Custom web fonts rendering jagged on 1080p standard DPI Windows desktop displays.
43	MEDIUM	Firefox Flex Gap Collapse	Outdated Firefox builds failing to calculate CSS gap properties in multi-line flex setups.
44	LOW	Middle-Click Autoscroll Lock	Clicking scroll-wheel inside containers with hidden overflow breaks auto-scrolling navigation.
45	HIGH	Windows High Contrast Mode	Accessibility mode removing borders, making outline buttons completely invisible on Windows.
46	LOW	Mouse Wheel Velocity Scaling	Smooth scroll libraries scrolling excessively fast when used with stepped physical mouse wheels.
47	LOW	SVG Gradient Missing in Edge	Complex linear gradient definitions inside SVG symbols disappearing in Chromium Edge.
48	MEDIUM	Print Stylesheet Export Distortion	Exporting web page to PDF/Print outputs unreadable dark backgrounds without print styles.
49	CRITICAL	Keyboard Focus Trap in Modals	Tab navigation getting trapped in background content while an inquiry popup is active.
50	HIGH	Missing Visible Focus Rings	Removing outline: none without :focus-visible fallbacks disables keyboard navigation.
51	LOW	Multi-Monitor DPI Drag Blurring	Dragging window from a 4K display to a 1080p display causes bitmap icon blurring.
52	LOW	Right-Click Context Menu Hijack	Custom mouse interaction handlers blocking standard copy/paste functionality on forms.
53	MEDIUM	Text Ellipsis Without Tooltip	Long medical terms truncated with . . . without hover tooltip explaining the full string.
54	MEDIUM	Enter Key Form Submission Glitch	Pressing Enter inside single-line inputs causing unexpected page refresh instead of action.
55	MEDIUM	Sticky Sidebar Overlapping Footer	Sticky navigation sidebar scrolling down directly over top of corporate footer content.
56	LOW	CSS Subgrid Incompatibility	Nested card alignments breaking on browser versions lacking native subgrid support.
57	HIGH		

CATEGORY 3: WINDOWS & CROSS-BROWSER DESKTOP BUGS (ITEMS 41 – 60)

#	SEVERITY	BUG / INSPECTION PARAMETER	TECHNICAL ROOT CAUSE & VERIFICATION POINT
		Blank Target Missing Rel Attributes	External links with <code>target="_blank"</code> missing <code>rel="noopener noreferrer"</code> security headers.
58	MEDIUM	Anchor Jump Hidden Behind Header	Clicking jump links (<code>#contact</code>) hides section title underneath sticky header (needs <code>scroll-margin-top</code>).
59	LOW	Hover State Stacking Overdraw	Multiple rapidly hovered grid cards lagging browser compositor thread.
60	MEDIUM	Video Codec Support Fallbacks	MP4 video tags lacking WebM fallback sources on open-source Linux/Chromium builds.

CATEGORY 4: NAVIGATION, LAYOUT & CONTENT STRUCTURE (ITEMS 61 – 80)

#	SEVERITY	BUG / INSPECTION PARAMETER	TECHNICAL ROOT CAUSE & VERIFICATION POINT
61	HIGH	Hero Title Duplication String	Concatenated page title repeating brand name twice in DOM and meta title headers.
62	CRITICAL	Broken Internal Route Links	Navigation header links pointing to dead '#' fragment targets or missing secondary pages.
63	MEDIUM	Tablet Breakpoint Grid Collision	Footer columns overlapping between 768px and 1024px viewport widths.
64	LOW	Breadcrumb Trail Disconnect	Subpages lacking semantic breadcrumb markup for navigational tracking.
65	MEDIUM	Card Height Asymmetry	Unequal text lengths causing staggered button alignments; requires <code>flex-col</code> with <code>mt-auto</code> .
66	LOW	Orphaned Single-Word Lines	H1 and H2 headings breaking with single dangling words; fix with <code>text-wrap: balance</code> .
67	HIGH	Corporate Logo Aspect Distortion	Brand logo image squished or stretched due to conflicting width/height CSS rules.
68	MEDIUM	Missing Multi-Resolution Favicons	404 responses for <code>favicon.ico</code> , <code>apple-touch-icon.png</code> , and Android icons.
69	MEDIUM	Mega-Menu Hover Tunnel Drop	Dropdown menus closing abruptly as cursor transitions across invisible padding boundaries.
70	HIGH	Language Switcher Desync	Toggling language switch changes layout state but fails to swap active translation strings.
71	CRITICAL	Legal & Impressum Entity Accuracy	Missing or mismatched corporate registration, VAT ID, or German Impressum requirements.
72	HIGH	Regulatory Badge Verification	EU-GMP and WHO-GMP compliance badges lacking formal descriptive labels and alt text.
73	HIGH	Unresponsive Pipeline Tables	R&D clinical pipeline data tables clipping and causing horizontal page overflow on mobile.
74	MEDIUM	Empty Search State Missing	

CATEGORY 4: NAVIGATION, LAYOUT & CONTENT STRUCTURE (ITEMS 61 – 80)

#	SEVERITY	BUG / INSPECTION PARAMETER	TECHNICAL ROOT CAUSE & VERIFICATION POINT
			Search and filter components displaying blank white space when zero query matches occur.
75	MEDIUM	Async Button Loading State	Submit button failing to display loading spinner or disabled state during network latency.
76	LOW	Inconsistent Corner Radii	Visual dissonance caused by mixing sharp square cards with rounded pills in identical sections.
77	HIGH	Low Contrast Ratios (WCAG AA)	Muted grey descriptive text failing minimum 4.5:1 contrast against light backgrounds.
78	MEDIUM	Backdrop Click Modal Dismissal	Clicking outside dialog overlay fails to close active modals, forcing explicit 'X' button tap.
79	MEDIUM	Multiple H1 Heading Tags	Multiple <h1> tags on a single page weakening technical SEO structure.
80	HIGH	Missing Alt Attributes on Images	Critical product and facility photography lacking accessible alt="" descriptions.

CATEGORY 5: PERFORMANCE, NETWORK, FORMS & SECURITY (ITEMS 81 – 105)

#	SEVERITY	BUG / INSPECTION PARAMETER	TECHNICAL ROOT CAUSE & VERIFICATION POINT
81	HIGH	Uncompressed Hero Assets (>2MB)	Serving raw PNG/JPG assets instead of next-generation optimized WebP/AVIF images.
82	CRITICAL	Silent Form Submission Failure	Inquiry form submission returning network errors without displaying clear user feedback banner.
83	HIGH	Client-Side Email Regex Flaws	Form accepting malformed email structures (e.g. missing top-level domain).
84	CRITICAL	XSS Sanitization Vulnerability	Contact form message inputs failing to sanitize special script characters on server payload.
85	HIGH	API Rate-Limiting Missing	Serverless contact endpoints vulnerable to automated bot form-submission spamming.
86	MEDIUM	Vercel Cache Header Headers	Static assets missing immutable cache-control headers, forcing repetitive server downloads.
87	HIGH	Render-Blocking Web Fonts	Synchronously imported Google Fonts causing Flash of Invisible Text (FOIT) on initial paint.
88	HIGH	Cumulative Layout Shift (CLS > 0.1)	Asynchronous banner injection shifting page content down after initial viewport render.
89	HIGH	Interaction to Next Paint (INP) Lag	Heavy JavaScript main-thread execution delaying button tap response beyond 200ms.
90	MEDIUM	Excessive Bundle Size Payload	Unused iconography and full lodash modules included in initial client JS bundle.
91	MEDIUM	OpenGraph Social Meta Missing	

CATEGORY 5: PERFORMANCE, NETWORK, FORMS & SECURITY (ITEMS 81 – 105)

#	SEVERITY	BUG / INSPECTION PARAMETER	TECHNICAL ROOT CAUSE & VERIFICATION POINT
			Missing og:image and og:description tags causing broken link previews on LinkedIn.
92	LOW	Twitter Card Tags Absent	Missing twitter:card tags resulting in plain text link rendering on X/Twitter.
93	HIGH	Missing Canonical Link Tag	Absence of <link rel="canonical"> causing SEO dilution across preview and production URLs.
94	CRITICAL	Robots.txt Production Disallow	Production environment retaining staging headers (Disallow: /), blocking search crawlers.
95	MEDIUM	Dynamic Sitemap.xml 404	Missing XML sitemap endpoint preventing indexing of nested therapeutic division pages.
96	HIGH	CORS Font Asset Blockage	Vercel CDN domain headers blocking font files due to missing Access-Control-Allow-Origin.
97	HIGH	Mixed Content Insecure Requests	Legacy static image URLs loading over insecure HTTP protocol inside HTTPS deployment.
98	CRITICAL	Missing Strict Security Headers	Missing Content-Security-Policy, X-Frame-Options, and X-Content-Type-Options.
99	CRITICAL	Pre-Consent Cookie Script Firing	Analytics tracking scripts firing before explicit user opt-in, violating EU GDPR mandates.
100	MEDIUM	Third-Party Script Thread Blocking	Unoptimized external tracking widgets degrading Core Web Vitals performance benchmarks.
101	HIGH	React useEffect Memory Leak	Window resize and scroll listeners remaining uncleaned upon component unmounting.
102	CRITICAL	Next.js Hydration Mismatch	Server-rendered HTML differing from client-rendered DOM nodes causing visual flicker and errors.
103	CRITICAL	Unhandled React Error Boundary Crash	Network API exceptions unmounting the entire application without a fallback error UI screen.
104	HIGH	Default 404 Error Page Fallback	Non-existent URLs displaying default unbranded Vercel error rather than custom 404 template.
105	HIGH	Production Source Map Exposure	Client-side production builds publishing full readable TypeScript source maps in DevTools.

3. Automated CI/CD Execution Command

To execute the automated multi-device Playwright test suite and generate an interactive test report, run the following commands in your project environment:

```
# 1. Install Playwright and all multi-OS browser binaries
npm install -D @playwright/test
npx playwright install --with-deps

# 2. Execute cross-platform audit against Vercel deployment
npx playwright test --project="Windows Chrome" --project="iOS Safari (iPhone 14)" --project="Android Chrome (Pixel 7)"

# 3. Open full HTML test result & failure report
npx playwright show-report
```